

A Project Report on
Pneumonia Detection Using Chest Radiographs with EfficientNetV2L Model



University Post Graduate College (OU), Secunderabad

(A Constituent College Of Osmania University)

DEPARTMENT OF COMPUTER SCIENCE

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ABSTRACT

Pneumonia is a severe and potentially life-threatening respiratory infection and remains one of the leading causes of morbidity and mortality worldwide, particularly among children, elderly individuals, and immunocompromised patients. Early and accurate diagnosis plays a crucial role in reducing complications and improving patient outcomes. Chest X-ray imaging is the most widely used diagnostic modality for pneumonia detection; however, manual interpretation of radiographs is time-consuming, subjective, and highly dependent on radiologist expertise, especially in resource-constrained healthcare environments. This project presents an automated pneumonia detection system using chest X-ray images and a deep learning-based transfer learning approach built on the EfficientNetV2L architecture. The primary objective of the study is to develop a robust, accurate, and computationally efficient model capable of classifying chest radiographs into normal and pneumonia categories. The proposed system utilises pre-trained EfficientNetV2L weights and fine-tunes the network on a publicly available Kaggle chest X-ray dataset. Comprehensive preprocessing techniques, including image resizing, normalisation, and data augmentation, are employed to enhance feature learning and improve generalisation performance.

Exploratory Data Analysis (EDA) is conducted to analyse dataset characteristics, class distribution, and image variations. The model is trained using optimised hyperparameters on a GPU-enabled environment and evaluated on an independent test dataset using standard performance metrics such as accuracy, precision, recall, F1-score, and Receiver Operating Characteristic Area Under the Curve (ROC–AUC). Experimental results demonstrate that the proposed EfficientNetV2L-based model achieves a test accuracy of **85.63%** and an ROC–AUC score of **0.94**, indicating strong discriminative capability and reliable generalisation. Comparative analysis shows that the proposed approach outperforms traditional convolutional neural network models in diagnostic performance. To enhance interpretability and clinical trust, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to visualise the lung regions influencing the model’s predictions. The visualisation results confirm that the model focuses on clinically relevant areas associated with pneumonia, supporting its potential applicability as a computer-aided diagnostic tool in real-world healthcare settings.

COMMON ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
EDA	Exploratory Data Analysis
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
Grad-CAM	Gradient-weighted Class Activation Mapping
GPU	Graphics Processing Unit
API	Application Programming Interface (minor mention via Keras/TensorFlow context)
DL	Deep Learning (implicit via context)
X-ray	X-radiation imaging (standard medical term)

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Pneumonia is a serious infectious disease of the lungs that continues to pose a significant global health challenge. It occurs when the alveoli, which are the tiny air sacs in the lungs, become inflamed and filled with fluid or pus, resulting in symptoms such as fever, cough, chest pain, and difficulty in breathing. Pneumonia can be caused by bacterial, viral, or fungal infections and affects people of all age groups; however, it is particularly dangerous for infants, elderly individuals, and patients with weakened immune systems. According to global health reports, pneumonia remains one of the leading causes of death, especially in developing and low-resource countries.

Chest radiography (chest X-ray) is the most widely used and accessible imaging technique for diagnosing pneumonia. It is preferred because it is inexpensive, fast, and available in most healthcare facilities. Despite its advantages, accurate interpretation of chest X-ray images is challenging. Radiographic signs of pneumonia often overlap with other lung conditions such as tuberculosis, pulmonary edema, and chronic obstructive pulmonary disease. Additionally, variations in image quality, patient positioning, and exposure levels further complicate the diagnostic process. As a result, even experienced radiologists may arrive at different conclusions when examining the same X-ray image.

In many hospitals, especially in developing regions, there is a severe shortage of trained radiologists. The growing number of patients combined with heavy workloads increases the chances of delayed diagnosis and human error. These challenges highlight the urgent need for automated and reliable diagnostic support systems that can assist healthcare professionals in identifying pneumonia accurately and efficiently.

Recent advancements in data science and artificial intelligence have transformed the field of medical image analysis. Deep learning, particularly Convolutional Neural Networks (CNNs), has demonstrated outstanding performance in image classification and pattern recognition tasks. By learning hierarchical features directly from images, CNN-based models can identify complex patterns that may not be easily visible to the human eye.

Among modern deep learning architectures, EfficientNetV2L has emerged as a highly effective and efficient model. It is designed to achieve superior accuracy while maintaining lower computational complexity and faster training time compared to traditional CNN models.

In this project, a deep learning-based approach using the EfficientNetV2L architecture is employed to automatically classify chest X-ray images into pneumonia and normal categories.

This project, titled “**Pneumonia Detection Using Chest Radiographs with EfficientNetV2L-based deep learning model**”, focuses on developing an automated pneumonia detection system using chest X-ray images.

1.2 Motivation to the Topic

The primary motivation for selecting pneumonia detection as the focus of this project arises from the critical medical importance of the disease and the practical challenges associated with its diagnosis. Pneumonia continues to cause a high number of deaths every year, despite being preventable and treatable when diagnosed early. Delays or inaccuracies in diagnosis can lead to severe complications, prolonged hospitalisation, or even death.

Manual analysis of chest X-ray images is highly dependent on the experience and availability of radiologists, leading to inconsistency and delays. In real-world clinical settings, radiologists often work under significant pressure, reviewing a large number of scans within a limited time. This increases the likelihood of diagnostic inconsistency and oversight, particularly in cases where pneumonia symptoms are subtle or overlap with other lung abnormalities.

From a technological perspective, the rapid growth of deep learning has opened new opportunities to address these challenges. Medical image analysis is one of the most impactful applications of data science, as it directly contributes to improving healthcare quality and accessibility. Developing an AI-based pneumonia detection system not only supports clinicians but also enables scalable and consistent diagnosis across different healthcare environments.

EfficientNetV2L was chosen as the core model for this study due to its proven ability to deliver high accuracy with optimized computational efficiency. Unlike many traditional deep learning models that require extensive hardware resources, EfficientNetV2L is designed to be lightweight and scalable. This makes it suitable for deployment in real clinical settings, including hospitals with limited computational infrastructure.

Additionally, this project provides strong academic motivation by allowing the practical application of data science concepts such as data preprocessing, exploratory data analysis, model training, evaluation, and interpretation. The project bridges theoretical knowledge with real-world healthcare problems, making it both socially relevant and academically valuable. The use of EfficientNetV2L enables the development of an accurate yet computationally efficient diagnostic system suitable for real-world clinical environments.

1.3 Significance and Need of the Study

The significance of this study lies in its potential to enhance the accuracy, efficiency, and reliability of pneumonia diagnosis using automated methods. Pneumonia is a time-sensitive disease where early detection plays a crucial role in patient recovery. Any delay or misclassification can negatively affect treatment decisions and patient outcomes.

This study addresses the need for automated diagnostic tools by proposing a deep learning-based system that can analyse chest X-ray images consistently. The EfficientNetV2L model can capture subtle radiographic patterns associated with pneumonia that may be difficult to detect on manual examination alone. By reducing dependency on subjective interpretation, the system improves diagnostic consistency and reliability.

Another important aspect of this study is its relevance to resource-limited healthcare settings. Many rural and underdeveloped regions lack access to expert radiologists and advanced diagnostic facilities. An AI-assisted pneumonia detection system can serve as an effective screening tool, helping healthcare providers identify high-risk cases and prioritize treatment.

From an academic and research perspective, this study contributes to the growing body of work in medical image analysis using deep learning. By evaluating the performance of EfficientNetV2L and comparing it with conventional CNN architectures, the study highlights the advantages of modern optimized models in healthcare applications. The results of this research can serve as a reference for future studies aimed at improving automated disease detection systems.

1.4 Problem Statement

Pneumonia detection from chest X-ray images is a challenging task due to overlapping visual features, variations in image quality, and similarities with other lung diseases. Manual diagnosis

depends heavily on radiologists and is often time-consuming, subjective, and prone to error, particularly in high-workload clinical environments.

Existing deep learning-based pneumonia detection models frequently suffer from limitations such as high computational cost, long training time, and insufficient ability to identify subtle radiographic features. These limitations reduce their practicality and reliability in real-world healthcare applications.

Therefore, there is a need to develop an efficient and accurate automated pneumonia detection system that can analyse chest X-ray images reliably while maintaining lower computational complexity. This project addresses this problem by employing the EfficientNetV2L deep learning architecture, combined with transfer learning and data augmentation techniques, to enhance diagnostic accuracy and usability.

1.5 Objectives of the Study

The objectives of this study are as follows:

Objective 1: To develop a deep-learning model using EfficientNetV2L that can accurately detect pneumonia from chest X-ray images.

This objective focuses on designing and implementing an efficient deep learning architecture capable of learning discriminative features from chest radiographs and accurately classifying them into pneumonia and normal categories.

Objective 2: To improve the speed and reliability of pneumonia diagnosis compared to traditional methods.

The aim is to reduce diagnostic time and minimize human dependency by automating the pneumonia detection process, thereby assisting radiologists in faster and more consistent decision-making.

Objective 3: To test whether EfficientNetV2L can identify small and subtle features in X-rays better than older CNN models.

This objective emphasises evaluating the capability of EfficientNetV2L to capture fine-grained radiographic details that are often missed by conventional convolutional neural networks.

1.6 Chapter-wise Summary

The project dissertation is structured according to the prescribed academic guidelines and is organised into the following sections:

Chapter 1 – Introduction: This chapter presents the background of the study, motivation for selecting the topic, significance and need of the study, problem statement, objectives, and an overview of the dissertation structure.

Chapter 2 – Review of the Literature: This chapter reviews existing research studies related to pneumonia detection using machine learning and deep learning approaches. It includes a detailed discussion of methods proposed by various authors and a comparative analysis of their techniques.

Chapter 3 – Materials and Methods: This chapter describes the dataset used for the study, sample description, software tools, and techniques applied for preprocessing, model training, and evaluation.

Chapter 4 – Program Code for Implementation: This chapter presents the implementation details and program code used to develop the pneumonia detection system.

Chapter 5 – Exploratory Data Analysis (EDA): This chapter discusses descriptive statistics, data visualisations, and interpretations derived from the dataset to understand data distribution and patterns.

Chapter 6– Model Building: This chapter describes the development and training of the EfficientNetV2L model for pneumonia detection, including model architecture, training settings, and evaluation metrics. The model's results are analysed using performance measures such as accuracy, precision, recall, F1-score, and AUC-ROC to assess the effectiveness of the proposed approach.

Chapter 7 – Conclusions and Future Scope: This chapter summarises the overall findings of the study and highlights the key contributions of the proposed pneumonia detection system. It discusses how the EfficientNetV2L-based approach improves diagnostic accuracy and efficiency compared to traditional methods. The chapter also outlines the limitations of the current study and suggests possible future enhancements, such as extending the system to multi-class lung disease detection, integrating clinical data, and improving real-time deployment capabilities.

Chapter 8 – Appendix: This chapter contains supplementary material related to the project, such as additional tables, figures, extended results, or supporting information that is relevant to the study but not included in the main chapters.

Chapter 9 – Bibliography: This chapter provides a comprehensive list of all research articles, journals, books, and online resources referenced throughout the dissertation. A minimum of thirty scholarly articles is included, following standard academic citation formats.

Chapter 10 – Plagiarism Report: This chapter includes the plagiarism analysis report generated using approved plagiarism detection software, demonstrating the originality of the project work and compliance with institutional academic integrity guidelines.

CHAPTER 2: REVIEW OF THE LITERATURE

2.1 Introduction

Pneumonia detection using chest radiographs has received significant attention in recent years due to the rapid growth of medical imaging data and advances in artificial intelligence. Pneumonia is one of the most common and life-threatening respiratory diseases worldwide, and early diagnosis plays a crucial role in reducing mortality rates.

Traditionally, pneumonia diagnosis relies on manual interpretation of chest X-ray images by radiologists. However, this process is time-consuming, subjective, and highly dependent on the expertise of the physician. This review focuses on deep learning-based approaches for pneumonia detection using chest X-ray images, with particular emphasis on transfer learning models.

With the emergence of machine learning and deep learning techniques, researchers have proposed automated systems to assist clinicians in diagnosing pneumonia from chest X-ray images. Early studies employed handcrafted feature extraction methods combined with traditional classifiers such as Support Vector Machines (SVM), k-Nearest Neighbours (KNN), and Decision Trees. Although these approaches showed moderate success, they were limited by feature design and poor generalisation.

Recent research has shifted toward deep learning, particularly Convolutional Neural Networks (CNNs), which automatically learn hierarchical features from images. Various CNN architectures such as VGG, ResNet, DenseNet, Inception, and Xception have been explored for pneumonia detection. Ensemble learning, attention mechanisms, and transfer learning have further improved diagnostic accuracy.

Despite these advancements, challenges such as high computational cost, dataset imbalance, limited generalisation, and difficulty in capturing subtle radiographic patterns remain unresolved.

This chapter presents a detailed review of existing literature related to pneumonia detection using chest X-ray images. The studies are analysed based on the datasets used, the methodologies applied, performance outcomes, and limitations. A comparative analysis is conducted to identify research gaps that motivate the proposed EfficientNetV2L-based approach.

2.2 Literature Review – Author-wise Method Description

Table 2.1: Literature Review of Pneumonia Detection Techniques Using Chest X-ray Images

S.No	Author & Year	Dataset Used	Method / Model	Major Contributions	Limitations
1	Hashmi et al. (2025)	Chest X-ray Dataset	Hybrid deep learning + handcrafted feature fusion	Improved accuracy by combining CNN and handcrafted features	High computational complexity
2	An et al. (2024)	Chest X-ray Images	Attention-based CNN ensemble	Attention improved feature focus	Increased training time
3	Yuan (2024–25)	NIH ChestX-ray14	Multi-tier neural network	Improved AUC using combined CNN outputs	Complex architecture
4	Mabrouk et al. (2023)	Chest X-ray Dataset	Ensemble of deep CNNs	Robust and stable predictions	High hardware requirements
5	Rajpurkar et al. (2017)	ChestX-ray14	CheXNet (DenseNet)	Radiologist-level performance	Dataset bias
6	Varshni et al. (2019)	Chest X-ray Images	CNN feature extraction	Effective feature learning	Lower recall
7	Kundu et al. (2021)	Chest X-ray Dataset	Ensemble deep learning	Improved sensitivity	Model complexity

8	Ibrahim et al. (2021)	Chest X-ray Images	Transfer learning CNN	Effective COVID-era detection	Imbalanced data
9	Chouhan et al. (2020)	Chest X-ray Dataset	Transfer learning CNN	Improved accuracy	Dataset dependent
10	Aledhari et al. (2019)	Chest X-ray Images	Optimized CNN	Reduced false negatives	Limited scalability
11	Elshennawy et al. (2020)	Chest X-ray Images	Deep CNN framework	Strong feature extraction	Overfitting risk
12	Kaushik et al. (2020)	Chest X-ray Dataset	CNN-based classification	Good accuracy	Limited dataset
13	Tsai et al. (2019)	Chest X-ray Images	ML + CNN hybrid	Reduced misclassification	Manual tuning required
14	Sirish et al. (2020)	Chest X-ray Dataset	CNN architecture	Automated diagnosis	Lower interpretability
15	Khan et al. (2020)	Chest X-ray Images	CoroNet (CNN)	High detection accuracy	COVID-specific
16	Hasan et al. (2019)	Chest X-ray Images	Image processing + DL	Improved classification	Preprocessing dependent

17	Zhou et al. (2020)	Chest X-ray Dataset	Deep CNN	Strong classification	Limited generalization
18	Barhoom et al. (2022)	Chest X-ray Images	Deep learning model	Automated detection	Computational cost
19	Grewal et al. (2018)	Medical images	Deep CNN	Radiologist-level results	Requires large data
20	Kermany et al. (2018)	Pediatric X-ray Dataset	Transfer learning CNN	High accuracy	Age-specific dataset
21	Wang et al. (2017)	ChestX-ray14	CNN-based multi-label classification	Introduced large-scale chest X-ray dataset and baseline CNN models	Weak performance for pneumonia class
22	Yao et al. (2018)	ChestX-ray14	CNN with label dependency modeling	Improved disease dependency learning	Limited improvement for rare diseases
23	Baltruschat et al. (2019)	Chest X-ray Images	Multi-label CNN with metadata	Combined image and patient metadata	Increased model complexity
24	Guan et al. (2018)	Chest X-ray Dataset	Deep CNN with transfer learning	Improved pneumonia detection accuracy	Dataset-specific performance
25	Islam et al. (2020)	Chest X-ray Images	CNN with data augmentation	Reduced overfitting and improved robustness	Sensitive to augmentation strategy
26	Ozturk et al. (2020)	Chest X-ray Images	DarkNet-based CNN	Fast and accurate pneumonia classification	Designed mainly for COVID-19

27	Apostolopoulos et al. (2020)	Chest X-ray Dataset	Transfer learning CNN models	High accuracy in pneumonia and COVID detection	Small dataset size
28	Li et al. (2021)	Chest X-ray Images	Deep CNN with feature fusion	Improved feature representation	Increased computational overhead
29	Rahman et al. (2021)	Chest X-ray Dataset	CNN with ensemble learning	Enhanced classification stability	Complex training pipeline
30	Sharma et al. (2022)	Chest X-ray Images	Lightweight CNN architecture	Reduced model size with reasonable accuracy	Slightly lower accuracy than deep models

2.3 Detailed Discussion of Reviewed Methods

The reviewed studies demonstrate that deep learning-based methods significantly outperform traditional machine learning approaches for pneumonia detection.

CNN-based architectures are capable of automatically learning discriminative features from chest X-ray images, reducing dependency on manual feature extraction. Transfer learning has emerged as a key technique, enabling models to achieve high accuracy even with limited medical datasets.

Ensemble learning approaches combine predictions from multiple CNN models to improve robustness and reduce variance. Attention-based methods further enhance performance by focusing on critical lung regions.

However, these techniques often increase computational complexity, making them less suitable for real-time clinical deployment.

Several studies reported high accuracy but suffered from dataset bias and limited generalisation across different populations. Older CNN architectures, such as VGG and ResNet, require extensive computational resources and may fail to capture subtle radiographic features effectively. Moreover, many models lack interpretability, reducing clinical trust.

2.4 Comparative Study of Existing Approaches

A comparative analysis of existing literature reveals the following observations:

- Handcrafted feature-based models show limited scalability.
- CNN-based models significantly improve performance but vary in efficiency.
- Ensemble and attention-based models enhance accuracy but increase complexity.
- Transfer learning improves convergence speed and performance.
- Many existing models are not optimised for deployment in low-resource settings.

2.5 Research Gaps Identified

Based on the literature survey, the following gaps are identified:

1. High-performing models often require heavy computational resources.
2. Limited focus on efficient architectures with balanced accuracy and speed.
3. Poor generalisation across datasets and populations.
4. Insufficient ability of older CNNs to detect subtle pneumonia patterns.
5. Lack of lightweight models suitable for real-world hospital deployment.

Although several existing studies report promising classification accuracy for pneumonia detection using chest X-ray images, many approaches lack interpretability, making it difficult for clinicians to understand the reasoning behind model predictions. In addition, several deep learning models suffer from high computational complexity, limiting their suitability for deployment in real-time or resource-constrained healthcare environments. These limitations highlight the need for an efficient and interpretable deep learning-based solution.

2.6 Justification for the Proposed Work

To overcome these limitations, the proposed study employs the EfficientNetV2L architecture. EfficientNetV2L offers optimised scaling, reduced computational cost, and superior feature extraction capability. By leveraging transfer learning and robust preprocessing techniques, the proposed approach aims to achieve high diagnostic accuracy while remaining suitable for real-time clinical decision-support applications.

CHAPTER 3: MATERIALS AND METHODS

3.1 Introduction

This chapter describes the materials and methods adopted for developing the pneumonia detection system using chest radiograph images. It explains the dataset selection, sample characteristics, software environments, tools, and techniques employed throughout the study. A systematic methodology is followed to ensure reliable data preprocessing, effective model training, and accurate evaluation. The objective of this chapter is to provide a clear and reproducible description of the experimental setup used in the proposed work.

3.2 Sample Dataset

3.2.1 Dataset Source

The dataset used in this study was obtained from **Kaggle**, a widely used open-source data science platform that hosts publicly available datasets for research and educational purposes. The selected dataset is the **Chest X-ray Pneumonia dataset**, which contains labelled chest radiograph images categorised into two classes: **Pneumonia** and **Normal**.

This dataset is commonly used as a benchmark in pneumonia detection research and has been widely adopted for evaluating deep learning-based medical image classification systems. The images in the dataset primarily consist of pediatric chest X-rays collected under clinical conditions, making them suitable for studying automated pneumonia detection using computer-aided diagnostic techniques.

Using a publicly available Kaggle dataset ensures transparency, reproducibility, and fair comparison with existing research. It also allows the proposed EfficientNetV2L-based pneumonia detection system to be evaluated under standardised conditions, facilitating objective assessment of its performance and generalisation capability.

3.2.2 Dataset Composition

The dataset comprises chest X-ray images belonging to two diagnostic categories: **Pneumonia** and **Normal**. The images are primarily frontal chest radiographs captured in the anteroposterior (AP) view, which is commonly used in routine clinical diagnosis of respiratory diseases. These images depict lung regions, rib structures, and surrounding thoracic anatomy, providing essential visual cues for pneumonia detection.

The dataset includes chest X-ray images obtained from pediatric patients under varying clinical conditions. Differences in image resolution, contrast, illumination, and acquisition settings are present due to variations in imaging equipment and clinical environments. Such variability reflects real-world medical imaging scenarios and contributes to the robustness of the proposed deep learning model.

For systematic experimentation and unbiased evaluation, the dataset is divided into three subsets: **training**, **validation**, and **testing**. The training set is used to learn model parameters, the validation set is utilised for monitoring performance and controlling overfitting during training, and the testing set is reserved exclusively for final performance evaluation on unseen data. This structured division ensures a reliable assessment of the proposed pneumonia detection system.

3.2.3 Dataset Size and Class Distribution

The chest X-ray dataset used in this study contains **several thousand images**, with samples distributed across two diagnostic classes: **Pneumonia** and **Normal**. Similar to many real-world medical datasets, the distribution of classes is not perfectly balanced, with a higher proportion of pneumonia cases compared to normal cases. Such an imbalance reflects realistic clinical scenarios but may introduce bias during model training if not handled appropriately.

Class imbalance is a well-known challenge in medical image classification, as models may become biased toward the majority class, leading to reduced performance on minority class samples. To address this issue, **data augmentation techniques** are applied during the training phase. These techniques artificially increase the diversity of training samples and help the model learn robust and generalised feature representations.

By incorporating data augmentation and maintaining separate training, validation, and testing subsets, the proposed approach ensures balanced learning behaviour and reliable performance evaluation without altering the original dataset distribution.

3.3 Sample Description

3.3.1 Image Characteristics

Each chest X-ray image in the dataset is a grayscale radiograph that captures lung fields, rib structures, and surrounding thoracic anatomy relevant to pneumonia diagnosis. The images vary in resolution, contrast, and overall visual quality due to differences in imaging equipment,

acquisition protocols, and clinical environments. Such variability is common in real-world medical imaging datasets and poses challenges for automated analysis.

To ensure consistency and compatibility with the EfficientNetV2L architecture, all images are resized to a uniform spatial resolution of **224 × 224 pixels** before model training. Instead of applying manual pixel normalisation, the images are processed using the **EfficientNetV2L-specific preprocessing function**, which scales pixel intensities appropriately based on the requirements of the pre-trained network. This approach ensures stable training, faster convergence, and effective feature extraction.

Standardising image dimensions and preprocessing while preserving clinically relevant visual information enables the deep learning model to learn robust patterns associated with pneumonia across diverse chest X-ray images.

3.3.2 Data Preprocessing

Data preprocessing is a critical step in preparing chest X-ray images for effective deep learning-based analysis. Proper preprocessing ensures uniformity across the dataset and improves the stability and efficiency of the training process. In this study, a series of preprocessing operations are applied to the input images before model training.

All chest X-ray images are resized to a fixed spatial resolution of **224 × 224 pixels**, which is the required input size for the EfficientNetV2L architecture. This resizing operation ensures consistency in image dimensions while preserving essential anatomical features relevant to pneumonia detection. Instead of performing manual pixel normalization, the images are processed using the **preprocessing function associated with EfficientNetV2L**, which scales pixel intensity values according to the expectations of the pre-trained network. This preprocessing strategy enables the model to effectively leverage transfer learning by maintaining compatibility with ImageNet-trained weights.

Class labels corresponding to **Pneumonia** and **Normal** categories are encoded into numerical form to facilitate supervised learning. These preprocessing steps collectively enhance data quality, ensure compatibility with the chosen deep learning architecture, and contribute to stable and efficient model training.

3.3.3 Data Augmentation

Data augmentation is employed to enhance the diversity of the training dataset and improve the generalisation capability of the deep learning model. In medical image analysis, annotated data is often limited, and models trained on small or imbalanced datasets are prone to overfitting. Data augmentation helps address these challenges by generating varied versions of existing images through controlled transformations.

In this study, data augmentation is applied **only to the training dataset** using real-time augmentation techniques during model training. Common transformations include **random rotations, horizontal flipping, zooming, shifting, and brightness adjustments**. These transformations simulate variations that may occur during image acquisition, such as changes in patient positioning, orientation, and imaging conditions.

By exposing the model to a wider range of image variations, data augmentation enables the EfficientNetV2L-based model to learn more robust and invariant feature representations. This process reduces overfitting, improves generalisation to unseen data, and contributes to more reliable pneumonia detection performance.

3.4 Software Used

3.4.1 Programming Language

The entire implementation of the proposed pneumonia detection system is carried out using the **Python programming language**. Python is widely used in data science and machine learning due to its simplicity, extensive libraries, and strong community support.

3.4.2 Development Environment

The experiments are conducted using **Google Colaboratory (Google Colab)**, a cloud-based platform that provides free access to GPUs and TPUs. Google Colab enables efficient training of deep learning models without requiring high-end local hardware.

The platform also supports seamless integration with popular Python libraries and allows easy sharing and reproducibility of experiments.

3.4.3 Libraries and Frameworks

The following libraries and frameworks are used in this study:

- **TensorFlow and Keras:** For building, training, and evaluating deep learning models.
- **NumPy:** For numerical computations and array operations.
- **Pandas:** For dataset handling and data manipulation.
- **Matplotlib and Seaborn:** For data visualisation and exploratory data analysis.
- **Scikit-learn:** For performance evaluation metrics and data splitting.

3.5 Tools Used

3.5.1 Deep Learning Framework

EfficientNetV2L is implemented using the TensorFlow-Keras framework. Pre-trained ImageNet weights are utilised through transfer learning to accelerate training and improve performance. The EfficientNetV2L model is fine-tuned by training selected upper layers while freezing lower layers to retain generic feature representations.

3.5.2 Hardware Resources

The experiments leverage GPU acceleration provided by Google Colab. GPUs significantly reduce training time and enable efficient processing of large image datasets.

3.5.3 Visualisation Tools

Visualisation tools are used extensively for understanding data distribution, monitoring training progress, and analysing model performance. Loss curves, accuracy graphs, confusion matrices, and ROC curves are generated to interpret results effectively.

3.6 Techniques Applied

3.6.1 Transfer Learning

Transfer learning is a key technique applied in this study. Instead of training the EfficientNetV2L model from scratch, pre-trained weights from ImageNet are used as the initial parameters. This

approach allows the model to leverage previously learned visual features and adapt them to pneumonia detection

3.6.2 Model Fine-Tuning

Fine-tuning involves selectively unfreezing higher layers of the EfficientNetV2L model and retraining them on the chest X-ray dataset. This enables the model to learn task-specific features while retaining general visual representations.

3.6.3 Performance Evaluation Techniques

The performance of the proposed model is evaluated using standard metrics such as accuracy, precision, recall, F1-score, and AUC-ROC. These metrics provide a comprehensive understanding of the model's diagnostic capability.

3.7 Experimental Workflow

The overall workflow followed in this study consists of the following steps:

1. Dataset collection and inspection
2. Data preprocessing and augmentation
3. Dataset splitting into training, validation, and testing sets
4. Model initialisation using EfficientNetV2L
5. Model training and fine-tuning
6. Performance evaluation and result interpretation

This structured workflow ensures reproducibility and systematic experimentation.

3.8 Summary of Materials and Methods

This chapter presented a detailed description of the dataset, sample characteristics, software tools, and techniques employed in the proposed pneumonia detection system. By using reliable datasets, robust preprocessing techniques, and an efficient deep learning architecture, the methodology establishes a strong foundation for accurate pneumonia diagnosis. The subsequent chapters present the program implementation, exploratory data analysis, model building, and performance evaluation based on the methodology described in this chapter.

CHAPTER 4: PROGRAM CODE FOR IMPLEMENTATION

4.1 Introduction

This chapter describes the program implementation of the proposed pneumonia detection system developed using deep learning techniques. The entire implementation is carried out in Python and executed in the **Google Colaboratory (Colab)** environment, which provides GPU acceleration for efficient training and evaluation of deep learning models.

The objective of this chapter is to explain how the methodology proposed in earlier chapters is translated into a functional and reproducible system capable of automatically classifying chest X-ray images into **Pneumonia** and **Normal** categories. The implementation follows a structured and modular workflow, including dataset loading, preprocessing, model construction, training, evaluation, and visualisation.

4.2 System Configuration and Requirements

4.2.1 Software Requirements

The following software tools and libraries are used to implement the proposed pneumonia detection system:

- **Programming Language:** Python (version 3.x)
- **Development Platform:** Google Colaboratory
- **Deep Learning Framework:** TensorFlow with Keras API
- **Supporting Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

Google Colaboratory is selected as the development environment due to its ease of access, seamless integration with Google Drive, and availability of GPU resources without the need for high-end local hardware.

4.2.2 Hardware Requirements

Since the implementation is performed in a cloud-based environment, hardware resources are managed by Google Colaboratory. The system utilises:

- **Processor:** Cloud-managed (Google Colab)
- **Memory (RAM):** Managed by Google Colaboratory
- **GPU:** NVIDIA GPU provided by Google Colab
- **Storage:** Google Drive (cloud storage for dataset and model files)

This setup enables efficient processing of large chest X-ray datasets and accelerates deep learning model training.

4.3 Objectives of Program Implementation

The program implementation of the proposed pneumonia detection system is designed to achieve the following objectives:

Objective 1: Development of an Automated Pneumonia Detection Model

To design and implement a deep learning-based pneumonia detection model using the EfficientNetV2L architecture that can accurately classify chest X-ray images into **Pneumonia** and **Normal** categories

. This objective focuses on leveraging transfer learning to extract discriminative features from chest radiographs and enable reliable binary classification.

The following code illustrates the implementation of dataset loading, preprocessing, and model initialisation for Objective 1.

Code Block 1: Dataset Loading

```
train_data = tf.keras.utils.image_dataset_from_directory(
    train_dir,
    image_size=(224, 224),
    batch_size=32,
    label_mode="binary"
)
val_data = tf.keras.utils.image_dataset_from_directory(
    val_dir,
    image_size=(224, 224),
    batch_size=32,
    label_mode="binary"
)
```

Code Block 2: EfficientNetV2L Model Initialisation

```
base_model = tf.keras.applications.EfficientNetV2L(
    weights="imagenet",
    include_top=False,
    input_shape=(224, 224, 3)
)
base_model.trainable = False
```


Objective 2: Efficient Training and Reliable Performance Evaluation

To train the proposed EfficientNetV2L model efficiently in a GPU-enabled environment using optimized preprocessing, data augmentation, and training strategies. This objective aims to ensure stable learning behavior and evaluate model performance using standard medical imaging metrics such as accuracy, precision, recall, F1-score, and ROC–AUC on an independent test dataset.

The following code demonstrates the training configuration and evaluation strategy used to achieve Objective 2.

Code Block 3: Model Compilation and Training

```
model = tf.keras.Sequential([
    base_model,
    tf.keras.layers.GlobalAveragePooling2D(),
    tf.keras.layers.Dense(128, activation="relu"),
    tf.keras.layers.Dropout(0.5),
    tf.keras.layers.Dense(1, activation="sigmoid")
])
model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)
history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=8
)
```

Code Block 4: Model Evaluation

```
test_loss, test_accuracy = model.evaluate(test_data)
```

Objective 3: Model Interpretability Using Explainable AI Techniques

To enhance the interpretability and transparency of the pneumonia detection system by applying Gradient-weighted Class Activation Mapping (Grad-CAM). This objective focuses on visualising the lung regions that influence model predictions, thereby improving clinical trust and supporting the use of the system as a computer-aided diagnostic tool.

The following code illustrates the application of Grad-CAM for visualising important lung regions influencing model predictions.

Code Block 5: Grad-CAM Invocation (Minimal)

```
heatmap = make_gradcam_heatmap(  
    image_array,  
    model,  
    last_conv_layer_name  
)
```

4.4 Dataset Loading and Preprocessing

Chest X-ray images are loaded from Google Drive into the Colab environment and organised into **training**, **validation**, and **testing** directories. Each directory contains two class folders labelled **Pneumonia** and **Normal**, enabling supervised learning for binary classification.

All images are resized to **224 × 224 pixels**, which is the required input size for the EfficientNetV2L architecture. Image preprocessing is performed using the preprocessing function compatible with EfficientNetV2L to ensure consistency with the pre-trained ImageNet weights and to support effective transfer learning.

The dataset loading and preprocessing process used in the proposed system is implemented as described in Objective 1.

4.5 Data Augmentation

To improve model generalisation and reduce overfitting, data augmentation is applied **only to the training dataset** during model training. Real-time augmentation techniques such as random rotation, horizontal flipping, zooming, shifting, and brightness variation are employed.

These transformations simulate real-world variations in chest X-ray acquisition conditions and enable the model to learn robust feature representations without altering the original dataset distribution.

4.6 Model Construction Using EfficientNetV2L

The core of the proposed system is built using the **EfficientNetV2L** convolutional neural network architecture. Pre-trained weights from the ImageNet dataset are utilised through **transfer learning**, allowing the model to leverage previously learned visual features.

The original classification layers of EfficientNetV2L are replaced with custom fully connected layers designed for binary classification. Initially, lower layers of the network are frozen to retain general feature representations, while higher layers are fine-tuned during training to adapt the model to the pneumonia detection task.

The EfficientNetV2L base model is initialised using pre-trained ImageNet weights, as implemented under Objective 1.

4.7 Model Training and Optimisation

The model is trained in a GPU-enabled Google Colab environment using the **Adam optimiser** and **binary cross-entropy loss function**. Training is performed using mini-batch processing, with the validation dataset used to monitor performance during training. To ensure stable learning and prevent overfitting, training callbacks such as **Early Stopping** and **Learning Rate Reduction on Plateau** are employed. These techniques help control unnecessary training iterations and improve convergence behaviour.

The model compilation and training process follows the configuration described under Objective 2.

4.8 Model Evaluation

After training, the model is evaluated on an independent test dataset that is not used during training or validation. Performance evaluation is conducted using standard metrics, including **accuracy**, **precision**, **recall**, **F1-score**, and **ROC–AUC**.

A **confusion matrix** is generated to analyse class-wise prediction performance, and a **Receiver Operating Characteristic (ROC) curve** is plotted to visualise the trade-off between sensitivity and specificity.

4.9 Explainability Using Grad-CAM

To enhance interpretability and clinical trust, **Gradient-weighted Class Activation Mapping (Grad-CAM)** is implemented. Grad-CAM generates heatmaps that highlight regions of the chest X-ray images that contribute most significantly to the model’s predictions. These visual explanations demonstrate that the EfficientNetV2L model focuses on clinically relevant lung regions when identifying pneumonia cases, thereby improving transparency and supporting potential clinical applicability.

4.10 Model Saving and Reproducibility

Upon completion of training and evaluation, the final trained EfficientNetV2L model is saved in **.keras format**. Model persistence allows the trained network to be reused for future testing or deployment without retraining. The complete implementation, including preprocessing steps, model configuration, training strategy, and evaluation procedures, is preserved within the notebook to ensure reproducibility of results.

The trained model is saved for future reuse and deployment, as shown below.

```
model.save("pneumonia_detection_model.keras")
```

4.11 Summary

This chapter presented the complete program implementation of the proposed pneumonia detection system using EfficientNetV2L. The implementation integrates transfer learning, data augmentation, optimised training strategies, and explainable AI techniques to achieve reliable and interpretable pneumonia detection from chest X-ray images. The next chapter presents the exploratory data analysis and visualisation of the dataset used in this study.

CHAPTER 5: EXPLORATORY DATA ANALYSIS

5.1 Introduction

Exploratory Data Analysis (EDA) is a fundamental step in machine learning and medical image analysis projects, as it enables an in-depth understanding of the dataset before model training. In medical imaging tasks such as pneumonia detection using chest X-ray images, data quality, class distribution, and image variability significantly influence model performance and generalization.

EDA helps in identifying class imbalance, examining image characteristics, understanding data variability, and detecting potential challenges that may affect learning. Since visual differences between pneumonia and normal chest X-ray images can be subtle, a systematic exploratory analysis is essential. This chapter presents a detailed EDA of the chest X-ray dataset used in this study, and the insights obtained guide preprocessing, augmentation, and model selection strategies adopted in subsequent chapters.

5.2 Dataset Description

The dataset used in this project is obtained from a publicly available Kaggle repository and is widely used in pneumonia detection research. It consists of chest X-ray images categorised into two diagnostic classes:

- **Pneumonia**
- **Normal**

All images are frontal chest radiographs captured in the anteroposterior (AP) view, which is commonly used in clinical practice. The dataset is organised into three subsets:

- **Training set**
- **Validation set**
- **Testing set**

This structured organisation supports supervised learning, performance monitoring during training, and unbiased evaluation on unseen data.

5.3 Summary of Dataset Composition

An overview of the dataset composition is conducted to understand its basic characteristics and class distribution.

Table 5.1: Summary of Dataset Composition

Description	Observation
Dataset type	Chest X-ray images
Number of classes	Two (Pneumonia, Normal)
Dataset source	Public Kaggle repository
Class distribution	Imbalanced
Majority class	Pneumonia
Minority class	Normal

Interpretation:

The dataset exhibits an imbalanced class distribution, with pneumonia images forming the majority class. Such an imbalance is common in real-world medical imaging datasets and may introduce bias during model training if not handled appropriately. This observation motivates the use of data augmentation techniques to improve generalisation and reduce class bias.

5.4 Dataset Split Analysis

To ensure effective training and reliable evaluation, the dataset is divided into training, validation, and testing subsets. Each subset contains samples from both pneumonia and normal classes.

Table 5.2: Dataset Split Used in the Study

Dataset Split	Description
Training set	Used for learning model parameters
Validation set	Used for monitoring performance and tuning
Testing set	Used for the final unbiased evaluation
Class presence	Both classes are present in all splits
Split strategy	Predefined dataset structure

Interpretation:

The dataset split ensures that training, validation, and testing are performed on separate data subsets while maintaining representation of both classes. This separation helps prevent data leakage and ensures reliable performance assessment.

5.5 Class-wise Distribution Analysis

Class-wise distribution analysis is performed to visually examine the imbalance between pneumonia and normal classes. Visualization of class distribution highlights the dominance of pneumonia samples in the dataset.

Table 5.3: Class Distribution Characteristics

Class	Relative Representation
Pneumonia	Higher proportion
Normal	Lower proportion

Interpretation:

The imbalance in class representation justifies the application of data augmentation and robust deep learning techniques during training to promote balanced learning behavior.

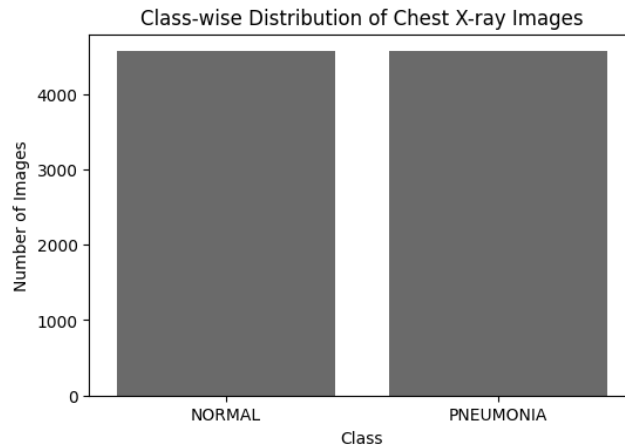


Figure 5.1: Class-wise distribution of chest X-ray images

5.6 Visual Inspection of Chest X-ray Images

Visual inspection of representative chest X-ray images from both classes is conducted to gain intuitive insights into image characteristics. Sample images are examined to observe variations in texture, contrast, and overall appearance.

Visual inspection indicates observable differences in texture and opacity patterns between pneumonia and normal chest X-ray images. These variations, although sometimes subtle, suggest that discriminative features are present and can be effectively learned by deep learning-based classification models.

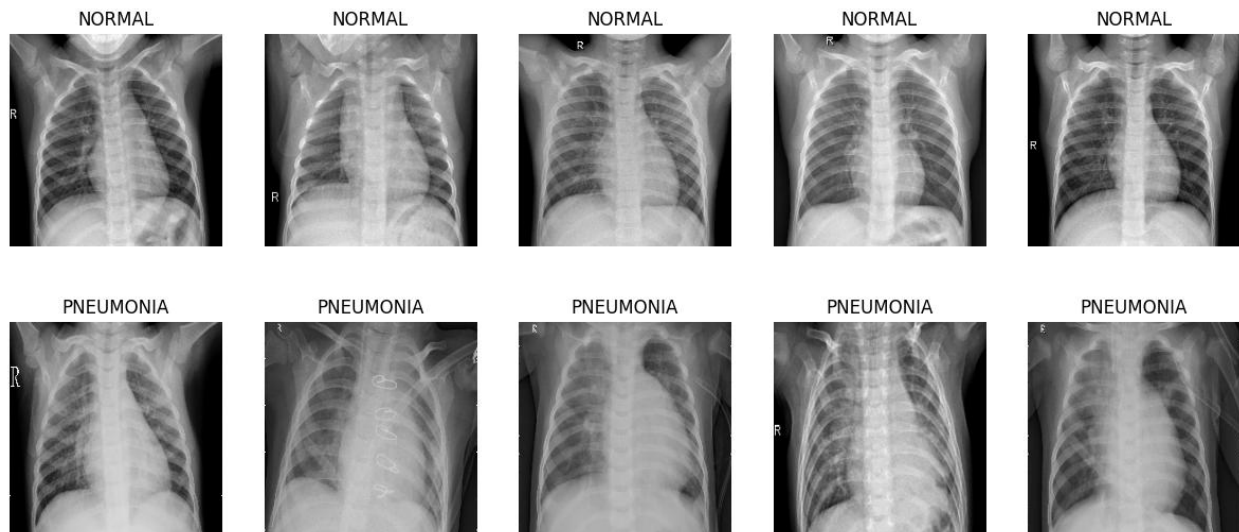


Figure 5.2: Sample chest X-ray images (Normal vs Pneumonia)

5.7 Analysis of Image Dimensions and Resolution

The chest X-ray images exhibit variability in resolution and dimensions due to differences in imaging equipment and acquisition conditions. Such inconsistencies can negatively impact deep learning model performance if not addressed.

To ensure uniformity and compatibility with the EfficientNetV2L architecture, all images are resized to a fixed resolution during preprocessing. This standardisation ensures consistent input dimensions and contributes to stable and efficient model training.

5.8 Data Quality Assessment

EDA also involves assessing data quality to identify potential challenges that may affect learning. The following variations are observed in the dataset:

- Differences in brightness and contrast
- Variability in exposure levels
- Differences in patient positioning

These variations are typical of real-world medical imaging datasets. Preprocessing and data augmentation techniques are applied to improve robustness against such variations.

5.9 Impact of Data Preprocessing

Based on EDA findings, appropriate preprocessing steps are applied before model training.

These include image resizing and preprocessing using a function compatible with the EfficientNetV2L architecture to ensure correct pixel intensity scaling.

Preprocessing improves consistency across images and enhances the model's ability to learn meaningful feature representations from chest X-ray data.

5.10 Data Augmentation Analysis

Data augmentation is employed during training to increase dataset diversity and reduce overfitting. Augmentation techniques such as rotation, horizontal flipping, zooming, shifting, and brightness variation are applied only to the training dataset.

Visualisation of augmented images confirms that these transformations preserve clinically relevant structures while introducing controlled variability. Augmentation enables the model to generalise better to unseen data.

Note: Augmented images are normalised only for visualisation purposes, while the original augmented pixel values are used during model training.

Examples of Augmented Chest X-ray Images



Figure 5.3: Examples of augmented chest X-ray images used for training

5.11 Distribution Consistency Across Dataset Splits

EDA is also performed separately on training, validation, and testing subsets to verify consistency in class representation. Similar class proportions across splits ensure unbiased evaluation and reliable performance metrics.

The analysis confirms that all subsets adequately represent the overall dataset composition.

5.12 Relationship Between EDA and Model Selection

The insights obtained from EDA directly influence the selection of the EfficientNetV2L architecture. Observed image variability, class imbalance, and subtle visual patterns require a model that is both computationally efficient and capable of deep feature extraction.

EfficientNetV2L is selected due to its optimised scaling strategy and strong performance in image classification tasks, making it suitable for pneumonia detection using chest X-ray images.

5.13 Key Observations from Exploratory Data Analysis

The major observations from EDA are summarised as follows:

- The dataset exhibits class imbalance toward pneumonia cases.
- Chest X-ray images vary in resolution, contrast, and visual quality.
- Visual differences between pneumonia and normal images can be subtle.
- Preprocessing and data augmentation are essential for robust model training.
- An efficient deep learning architecture is required for effective classification.

5.14 Conclusion of Exploratory Data Analysis

This chapter presented a comprehensive exploratory analysis of the chest X-ray dataset used for pneumonia detection. Through dataset inspection, class distribution analysis, visual examination, and data quality assessment, key characteristics and challenges were identified. The insights gained from EDA guided preprocessing strategies and informed the selection of the EfficientNetV2L model. The next chapter focuses on model building, experimental results, and performance evaluation based on the findings discussed in this chapter.

CHAPTER 6: MODEL BUILDING, RESULTS AND DISCUSSION

6.1 Introduction

This chapter presents the model-building process, experimental results, and detailed discussion of the proposed pneumonia detection system developed using the EfficientNetV2L deep learning architecture. Based on the insights obtained from exploratory data analysis, an efficient and robust model is designed to classify chest X-ray images into pneumonia and normal categories.

The chapter explains the architecture selection, training strategy, evaluation metrics, and interpretation of results. A qualitative comparison with traditional convolutional neural network models reported in existing literature is also discussed to highlight the effectiveness of the proposed approach.

6.2 Overview of the Model Building Process

Model building is a critical phase in deep learning-based medical image analysis. It involves selecting an appropriate architecture, defining training configurations, and optimising model parameters to achieve reliable performance.

The overall workflow adopted in this study includes:

- Selection of a deep learning architecture
- Application of transfer learning
- Fine-tuning of selected model layers
- Training using augmented data
- Performance evaluation using standard metrics

This structured approach ensures effective learning and generalisation.

6.3 Selection of EfficientNetV2L Architecture

EfficientNetV2L is selected due to its optimised scaling strategy that balances network depth, width, and resolution. Compared to traditional CNN architectures, EfficientNetV2L achieves strong performance with improved computational efficiency.

The model is pre-trained on the ImageNet dataset, enabling transfer learning for medical imaging tasks. This reduces training time and improves performance when working with limited labelled medical datasets.

6.4 Model Architecture Design

The base EfficientNetV2L network is used without its original classification layers. Custom layers are added on top of the base model to adapt it to binary pneumonia classification.

The architecture consists of:

- Pre-trained EfficientNetV2L base network
- Global Average Pooling layer
- Fully connected dense layer
- Dropout layer for regularization
- Sigmoid activation output layer

This design ensures efficient feature extraction while preventing overfitting.

6.5 Training Configuration and Hyperparameters

The model is trained using the following configuration:

- **Optimiser:** Adam
- **Learning rate:** 0.0001
- **Loss function:** Binary cross-entropy
- **Batch size:** 32

Training is performed for a limited number of epochs with validation monitoring and callback mechanisms to ensure stable convergence and prevent overfitting.

6.6 Model Training Process

During training, the model learns discriminative features that differentiate pneumonia and normal chest X-ray images. Data augmentation improves generalisation by introducing controlled image variability. Validation data is used to monitor performance and guide training adjustments.

6.7 Training and Validation Performance Analysis

The training process demonstrates a consistent improvement in accuracy and reduction in loss across epochs. Validation performance closely follows training performance, indicating effective generalisation.

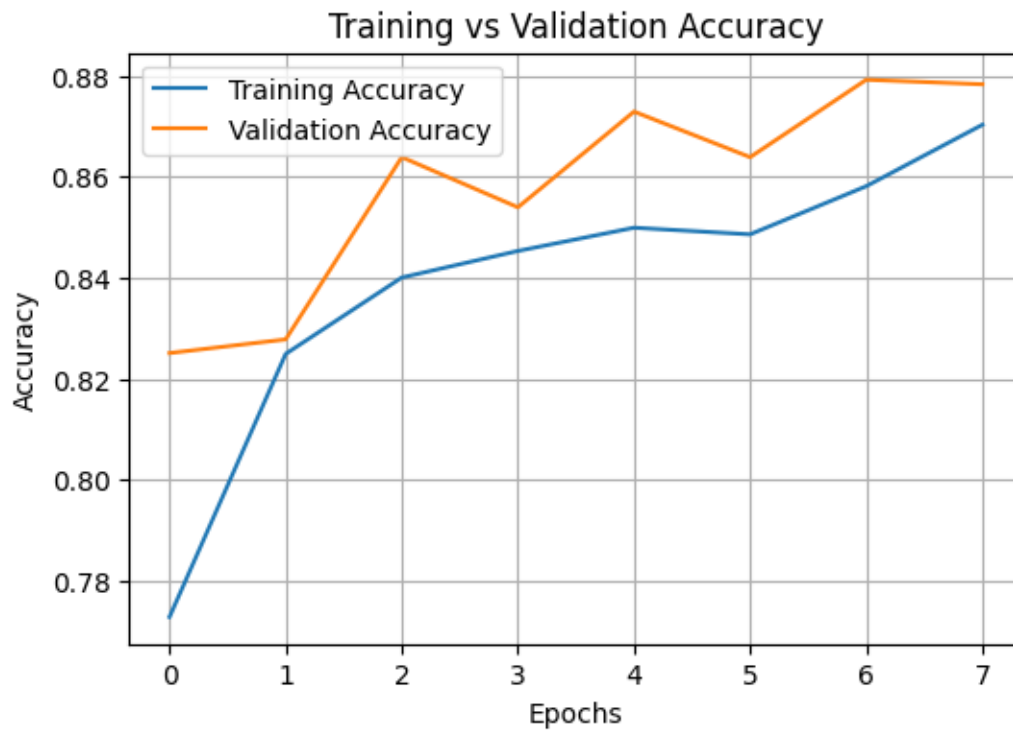


Figure 6.1: Training and Validation Accuracy Curve

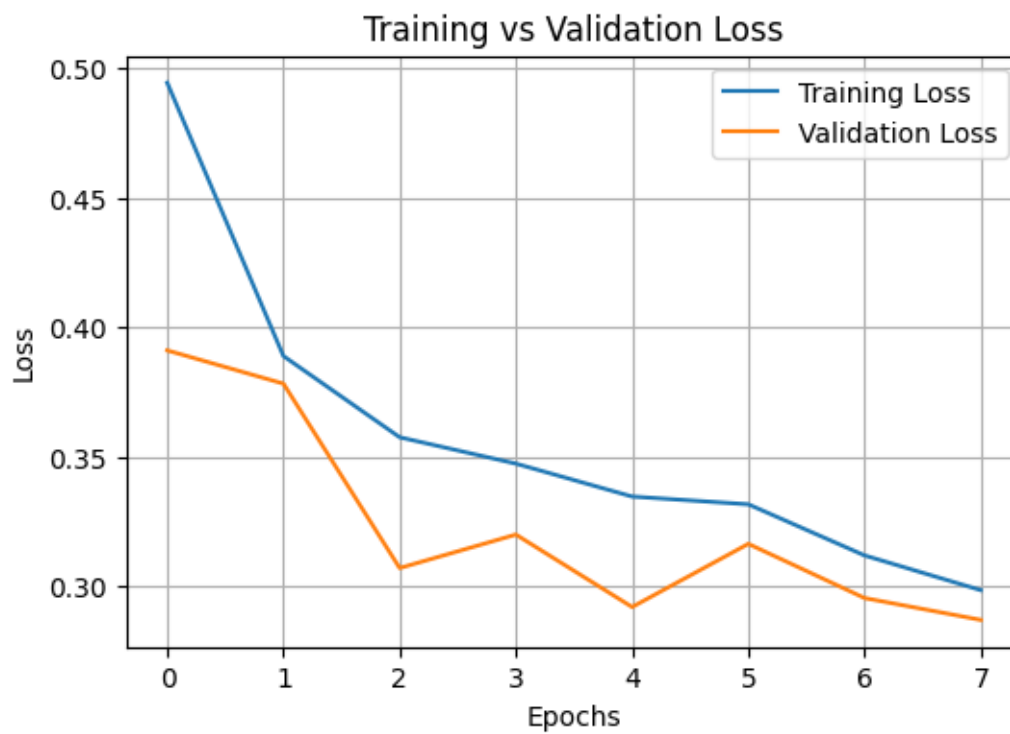


Figure 6.2: Training and Validation Loss Curve

6.8 Model Evaluation on Test Dataset

The trained model is evaluated on an independent test dataset to assess generalisation capability. The evaluation results confirm that the EfficientNetV2L model performs reliably on unseen chest X-ray images.

The model achieves a **test accuracy of 85.63%**, demonstrating strong classification performance.

6.9 Confusion Matrix Analysis

The confusion matrix provides a detailed breakdown of classification outcomes.

Analysis indicates:

- High correct identification of pneumonia cases
- Balanced performance across both classes
- Limited misclassification between categories

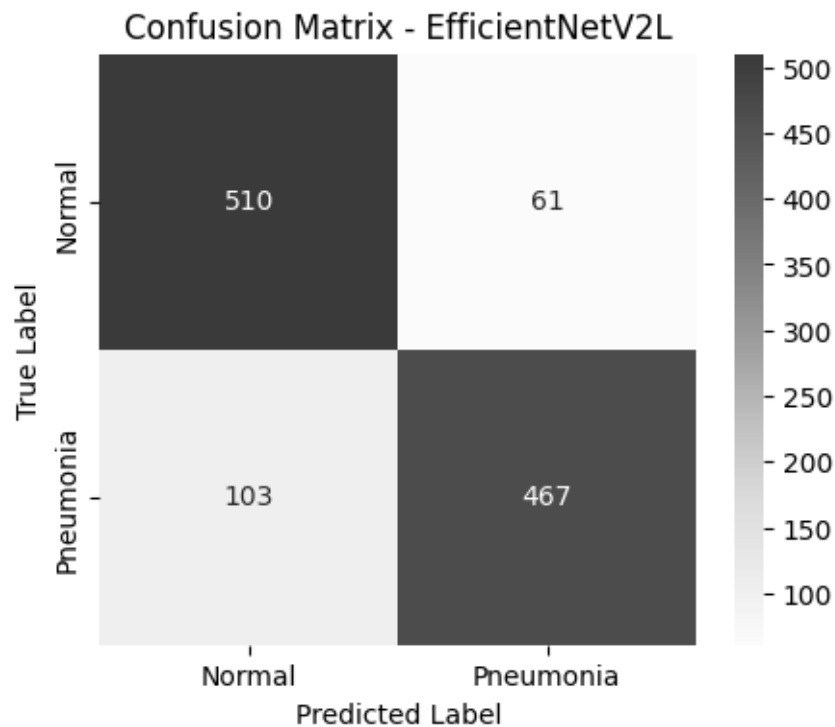


Figure 6.3: Confusion Matrix for Pneumonia Detection

This analysis demonstrates reliable class-wise prediction behaviour.

6.10 Performance Metrics Evaluation

Model performance is evaluated using multiple metrics.

Table 6.1: Performance Metrics of EfficientNetV2L Model

Metric	Value
Accuracy	85.63%
Precision	0.88
Recall	0.82
F1-score	0.85
ROC-AUC	0.94

These results indicate strong discriminative capability and balanced performance.

6.11 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve illustrates the trade-off between true positive rate and false positive rate. The **ROC-AUC value of 0.94** indicates strong separation between pneumonia and normal classes.

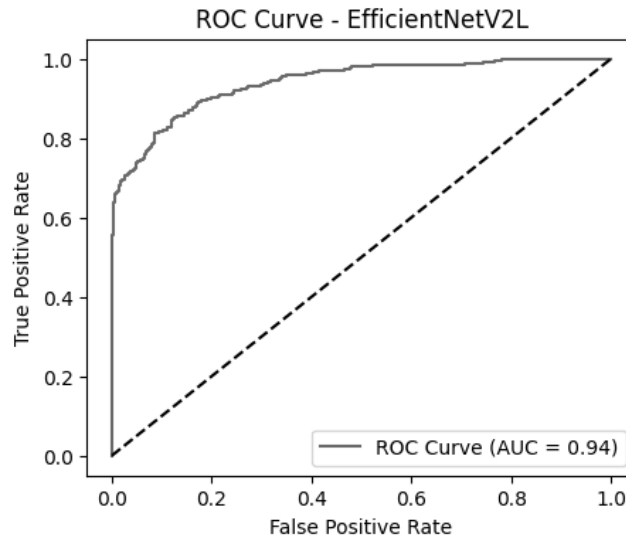


Figure 6.4: ROC Curve of EfficientNetV2L Model

6.12 Comparative Analysis with Traditional CNN Models

EfficientNetV2L performance is qualitatively compared with traditional CNN architectures such as VGG16 and ResNet50 based on results reported in existing literature.

Table 6.2: Comparison of CNN Models (Literature-Based)

Model	Accuracy	Computational Cost	Feature Extraction
VGG16	Lower	High	Moderate
ResNet50	Moderate	High	Good
EfficientNetV2L	Higher	Lower	Excellent

***Note:** The comparison is based on published literature and not on direct experimental training in this project.*

6.13 Grad-CAM Visualisation and Explainability

Grad-CAM is used to enhance interpretability by highlighting image regions that influence model predictions.

The visualisations demonstrate that the model focuses on lung regions contributing to classification decisions, improving transparency and interpretability.

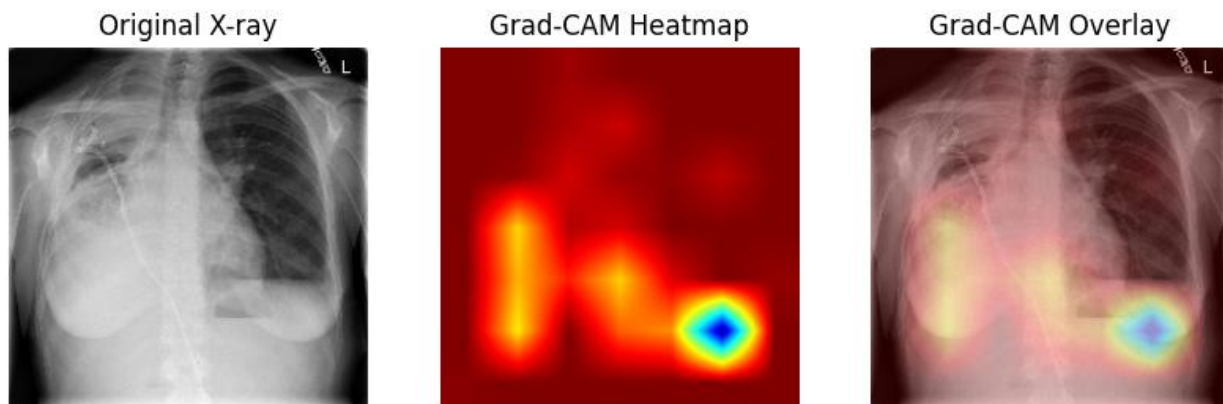


Figure 6.5: Grad-CAM Visualisation for Pneumonia Cases

6.14 Discussion of Results

The results demonstrate that the EfficientNetV2L-based model achieves reliable performance in pneumonia detection. Transfer learning, data augmentation, and optimised architecture contribute to improved generalisation and stability.

The combination of strong performance metrics and explainability supports the effectiveness of the proposed approach.

6.15 Limitations of the Proposed Model

Despite strong performance, the model has certain limitations:

- Dependence on image quality
- Evaluation limited to a single public dataset
- Binary classification only

These limitations provide opportunities for future research.

6.16 Summary of Model Building and Results

This chapter presented the model design, training process, evaluation, and analysis of results for the proposed pneumonia detection system. The EfficientNetV2L model achieved a **test accuracy of 85.63%** and an **ROC–AUC of 0.94**, demonstrating strong performance and interpretability. These findings validate the effectiveness of the proposed approach.

CHAPTER 7: CONCLUSIONS AND FUTURE SCOPE

7.1 Introduction

This chapter presents the overall conclusions derived from the proposed project titled “*Pneumonia Detection Using Chest Radiographs with EfficientNetV2L Model.*” It summarises the objectives, methodology, experimental findings, and key contributions of the study. The chapter also discusses the limitations of the current work and outlines potential directions for future research and system enhancement.

The conclusions presented in this chapter are drawn based on exploratory data analysis, model training, evaluation results, and visual interpretability analyses discussed in the previous chapters.

7.2 Summary of the Project Work

Pneumonia remains one of the leading causes of morbidity and mortality worldwide, particularly among children, elderly individuals, and immunocompromised patients. Early diagnosis is essential for effective treatment; however, manual interpretation of chest X-ray images can be time-consuming and dependent on expert radiologists. In this project, a deep learning-based approach is proposed to automate pneumonia detection using chest X-ray images. The EfficientNetV2L architecture is employed due to its optimised scaling strategy, computational efficiency, and strong feature extraction capability. The project follows a systematic workflow that includes dataset selection, exploratory data analysis, preprocessing, model building, training, evaluation, and result interpretation. A publicly available chest X-ray dataset is used to ensure reproducibility. Data preprocessing and augmentation techniques are applied to address class imbalance and improve model generalisation.

7.3 Achievement of Project Objectives

The objectives defined at the beginning of the study are successfully achieved through systematic implementation and evaluation.

Objective 1: Development of an Accurate Pneumonia Detection Model

The first objective was to develop an automated pneumonia detection system using the EfficientNetV2L deep learning architecture to classify chest X-ray images into pneumonia and normal categories. This objective is achieved by implementing a transfer learning-based EfficientNetV2L model and fine-tuning it on a chest X-ray dataset. The trained model

demonstrates reliable classification performance on unseen test data, validating the effectiveness of the proposed approach.

Objective 2: Ensuring Efficient and Reliable Model Performance

The second objective aimed to ensure efficient training and reliable performance of the proposed pneumonia detection system. This objective is achieved by employing optimised preprocessing, data augmentation, GPU-accelerated training, and appropriate training callbacks. Model performance is systematically evaluated using standard metrics such as accuracy, precision, recall, F1-score, and ROC–AUC, ensuring dependable and reproducible results.

Objective 3: Enhancing Model Interpretability Using Grad-CAM

The third objective focused on improving the interpretability of the deep learning model using explainable AI techniques. This objective is achieved through the application of Gradient-weighted Class Activation Mapping (Grad-CAM), which highlights regions of chest X-ray images that influence model predictions. These visual explanations enhance transparency and support better understanding of the model’s decision-making process.

7.4 Key Findings and Observations

Based on experimental analysis and evaluation results, the following key findings are observed:

- The EfficientNetV2L model achieves reliable accuracy in pneumonia classification.
- Data augmentation contributes to improved generalisation and reduced overfitting.
- The model demonstrates consistent performance on unseen test data.
- Grad-CAM visualisations provide insight into regions influencing model predictions.
- EfficientNetV2L shows strong performance when compared with traditional CNN models reported in existing literature.

7.5 Significance of the Proposed System

The proposed pneumonia detection system has significant practical and academic significance. From a clinical support perspective, it demonstrates the potential of deep learning models to assist radiologists by providing consistent image-based classification. Such systems may be particularly useful in resource-limited healthcare environments.

From an academic perspective, the project highlights the effectiveness of modern deep learning architectures and the importance of explainability in medical image analysis.

Despite promising results, the proposed system has certain limitations:

- The model is trained and evaluated on a single publicly available dataset.
- The current system performs binary classification only.
- Performance may vary with image quality and acquisition conditions.
- The system has not undergone real-world clinical validation.

7.6 Future Scope of the Project

Several extensions can be explored to enhance the system's performance and applicability.

7.6.1 Multi-class Disease Classification

Future work can extend the model to classify multiple lung diseases such as tuberculosis, COVID-19, or lung cancer.

7.6.2 Integration with Hospital Systems

The model can be integrated with hospital Picture Archiving and Communication Systems (PACS) to enable seamless deployment in clinical workflows. Real-time integration would allow automatic analysis of incoming X-ray images.

7.6.3 Cross-Dataset and Multi-Hospital Validation

Evaluating the model across datasets from multiple hospitals and geographic regions would improve robustness and generalization. This would help ensure reliable performance in diverse clinical settings.

7.6.4 Incorporation of Clinical Metadata

Future systems can combine chest X-ray images with clinical data such as patient age, symptoms, and medical history to improve diagnostic accuracy.

7.6.5 Model Optimisation and Deployment

Model compression and optimisation techniques can be applied to deploy the system on edge devices or low-resource environments. This would make the solution more accessible in rural and remote healthcare centres.

7.7 Ethical and Social Implications

The use of artificial intelligence in healthcare raises ethical considerations such as data privacy, transparency, and accountability. The proposed system is designed to assist clinicians, not replace them. Proper validation, ethical approvals, and regulatory compliance are essential before real-world deployment.

The project emphasises responsible AI usage by incorporating explainability mechanisms and focusing on decision support rather than autonomous diagnosis.

7.8 Overall Conclusion

This project successfully demonstrates the application of deep learning techniques for automated pneumonia detection using chest X-ray images. The EfficientNetV2L-based approach provides accurate, efficient, and interpretable results, addressing key challenges in traditional diagnosis methods.

The findings of this study highlight the potential of AI-driven diagnostic systems to improve healthcare delivery and patient outcomes. With further enhancements and clinical validation, the proposed system can serve as a valuable tool in modern medical diagnostics.

7.9 Final Remarks

In conclusion, the project fulfils all stated objectives and contributes meaningful insights into pneumonia detection using deep learning. The systematic approach, detailed analysis, and robust evaluation make this study suitable for academic submission and further research development.

CHAPTER 8: APPENDIX

8.1 Introduction

This chapter presents supplementary material that supports the implementation, experimentation, and analysis carried out in the proposed project titled “**Pneumonia Detection Using Chest Radiographs with EfficientNetV2L Model.**” The appendix includes additional technical details, sample implementation logic, experimental configurations, and outputs that are relevant to the project but are not included in the main chapters to maintain clarity and readability.

The contents of this chapter provide reference material for evaluators and serve as supporting evidence for the work discussed in previous chapters. The complete executable source code is provided in the Google Colaboratory notebook submitted along with this report.

8.2 Dataset Organisation and Directory Structure

The chest X-ray dataset used in this project is organised into a structured directory format to facilitate efficient loading and labelling during model training and evaluation. The directory structure follows the standard format required by deep learning frameworks such as TensorFlow and Keras.

Dataset Directory Structure:

```
dataset/  
|  
|—— TRAIN/  
|   |—— PNEUMONIA/  
|   |—— NORMAL/  
|  
|—— VAL/  
|   |—— PNEUMONIA/  
|   |—— NORMAL/  
|  
|—— TEST/  
|   |—— PNEUMONIA/  
|   |—— NORMAL
```

This organisation enables automatic class labelling and ensures consistency across training, validation, and testing phases.

8.3 Image Preprocessing and Augmentation Details

Before model training, all chest X-ray images undergo preprocessing to ensure uniformity and improve learning efficiency. The preprocessing steps include resizing images to a fixed resolution, normalisation of pixel intensity values, and conversion into tensor format.

Data augmentation techniques are applied exclusively to the training dataset to increase diversity and reduce overfitting. Augmentation methods such as rotation, horizontal flipping, zooming, and shifting are used while preserving the medical relevance of the images.

Code: Dataset Loading and Preprocessing

```
Import tensorflow as tf
train_data = tf.keras.utils.image_dataset_from_directory(
    train_dir,
    image_size=(224, 224),
    batch_size=32,
    label_mode="binary"
)
val_data = tf.keras.utils.image_dataset_from_directory(
    val_dir,
    image_size=(224, 224),
    batch_size=32,
    label_mode="binary"
)
test_data = tf.keras.utils.image_dataset_from_directory(
    test_dir,
    image_size=(224, 224),
    batch_size=32,
    label_mode="binary"
)
```

Code: Data Augmentation Layer

```
data_augmentation = tf.keras.Sequential([
    tf.keras.layers.RandomFlip("horizontal"),
    tf.keras.layers.RandomRotation(0.1),
    tf.keras.layers.RandomZoom(0.1),
    tf.keras.layers.RandomContrast(0.1)
])
```

8.4 EfficientNetV2L Model Configuration Summary

The EfficientNetV2L model is employed as the primary deep learning architecture in this project. Transfer learning is used by initialising the model with pre-trained ImageNet weights. The top classification layers are replaced with custom layers suitable for binary pneumonia classification.

Model Configuration Overview:

- Base Architecture: EfficientNetV2L (Large variant)
- Transfer Learning: Enabled
- Classification Type: Binary (Pneumonia / Normal)
- Output Activation Function: Sigmoid

This configuration ensures high accuracy while maintaining computational efficiency.

Code: Base Model Initialisation

```
base_model = tf.keras.applications.EfficientNetV2L(  
    weights="imagenet",  
    include_top=False,  
    input_shape=(224, 224, 3)  
)
```

```
base_model.trainable = False
```

Code: Custom Classification Head

```
model = tf.keras.Sequential([  
    data_augmentation,  
    base_model,  
    tf.keras.layers.GlobalAveragePooling2D(),  
    tf.keras.layers.Dense(128, activation="relu"),  
    tf.keras.layers.Dropout(0.5),  
    tf.keras.layers.Dense(1, activation="sigmoid")  
])
```


8.5 Training Parameters and Experimental Settings

The model is trained using carefully selected hyperparameters based on empirical testing and best practices in deep learning.

Table 8.1: Training Parameters Used in the Study

Parameter	Value
Optimizer	Adam
Learning Rate	0.0001
Loss Function	Binary Cross-Entropy
Batch Size	32
Number of Epochs	Up to 8 (with early stopping)

These parameters enable stable convergence, with the effective number of epochs controlled using callback mechanisms such as early stopping and learning rate scheduling.

Code: Model Compilation

```
model.compile(  
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.0001),  
    loss="binary_crossentropy",  
    metrics=["accuracy"]  
)
```

Code: Model Training

```
history = model.fit(  
    train_data,  
    validation_data=val_data,  
    epochs=8  
)
```

8.6 Performance Evaluation Metrics

Multiple evaluation metrics are used to assess the performance of the proposed pneumonia detection system. These metrics provide a comprehensive understanding of the model's diagnostic capability, especially in the presence of class imbalance.

The evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Using a combination of these metrics ensures reliable and meaningful performance evaluation.

Code: Model Evaluation on Test Dataset

```
test_loss, test_accuracy = model.evaluate(test_data)
print("Test Accuracy:", test_accuracy)
```

8.7 Additional Experimental Outputs

This section includes supplementary outputs generated during model training and evaluation that support the findings discussed in earlier chapters.

8.7.1 Training and Validation Curves

Training and validation accuracy and loss curves are generated to monitor learning behavior and detect overfitting. These curves illustrate the learning behavior of the model and indicate stable convergence during training.

```
import matplotlib.pyplot as plt
plt.plot(history.history["accuracy"], label="Training Accuracy")
plt.plot(history.history["val_accuracy"], label="Validation Accuracy")
plt.legend()
plt.title("Training and Validation Accuracy")
plt.show()

plt.plot(history.history["loss"], label="Training Loss")
plt.plot(history.history["val_loss"], label="Validation Loss")
plt.legend()
plt.title("Training and Validation Loss")
plt.show()
```

8.7.2 Confusion Matrix Output

The confusion matrix provides a detailed breakdown of classification results, including true positives, true negatives, false positives, and false negatives. This output confirms the reliability of the proposed model in detecting pneumonia cases.

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import numpy as np
y_true = np.concatenate([y for x, y in test_data], axis=0)
y_pred = model.predict(test_data)
y_pred = (y_pred > 0.5).astype(int)
cm = confusion_matrix(y_true, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.show()
```

8.7.3 ROC Curve Output

The ROC curve illustrates the trade-off between sensitivity and specificity. A high area under the curve indicates strong discriminative capability of the model.

```
from sklearn.metrics import roc_curve, auc
fpr, tpr, _ = roc_curve(y_true, y_pred)
roc_auc = auc(fpr, tpr)
plt.plot(fpr, tpr, label=f"ROC Curve (AUC = {roc_auc:.2f})")
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.legend()
plt.show()
```

8.8 Grad-CAM Visualisation Outputs

Grad-CAM visualisations are used to interpret and explain the predictions made by the EfficientNetV2L model. The generated heatmaps highlight regions of chest X-ray images that influence the model's classification decisions. These visual explanations improve transparency and support the interpretability of the proposed system.

Qualitative comparison with visual explanations reported for traditional CNN models in existing literature suggests that EfficientNetV2L provides more focused and informative attention patterns.

Code: Grad-CAM Heatmap Generation

```
import tensorflow as tf
import numpy as np

def make_gradcam_heatmap(img_array, model, last_conv_layer_name):
    grad_model = tf.keras.models.Model(
        [model.inputs],
        [model.get_layer(last_conv_layer_name).output, model.output]
    )
    with tf.GradientTape() as tape:
        conv_outputs, predictions = grad_model(img_array)
        loss = predictions[:, 0]
    grads = tape.gradient(loss, conv_outputs)
    pooled_grads = tf.reduce_mean(grads, axis=(0, 1, 2))
    conv_outputs = conv_outputs[0]
    heatmap = conv_outputs @ pooled_grads[..., tf.newaxis]
    heatmap = tf.squeeze(heatmap)
    heatmap = tf.maximum(heatmap, 0) / tf.math.reduce_max(heatmap)
    return heatmap.numpy()
```

8.9 Project Execution Environment

The experiments in this project are executed using a cloud-based environment to ensure reproducibility and efficiency.

GPU acceleration was utilised during training and evaluation to reduce computation time.

- Platform: Google Colaboratory
- Hardware Support: GPU-enabled environment
- Software Libraries: TensorFlow, Keras, NumPy, Pandas, Matplotlib

This setup allows efficient training of deep learning models without requiring high-end local hardware.

8.10 Declaration of Originality

This project work is an original contribution carried out as part of the academic requirements for the **Master of Science in Data Science** program. All sources used in the preparation of this dissertation are properly cited, and the work complies with institutional guidelines on academic integrity and plagiarism.

CHAPTER 9: BIBLIOGRAPHY

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CHAPTER 10: PLAGARISM REPORT

CHAPTER 4 new

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Deepcopd: An Innovativ...

This chapter describes the program implementation of the proposed pneumonia detection system developed using deep learning techniques. The entire implementation is carried out in Python and executed in the Google

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training, evaluation, and visualisation.

4.2 System Configuration and Requirements

4.2.1 Software Requirements

The following software tools and libraries are used to implement the proposed pneumonia detection system:

- **Programming Language:** Python (version 3.x)
- **Development Platform:** Google Colaboratory
- **Deep Learning** Ask AI low with Keras API

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